SULIT

Fakulti Komputeran

UNIVERSITI TEKNOLOGI MALAYSIA
FINAL EXAMINATION SEMESTER II 2022 / 2023

SUBJECT CODE : **SECR2043 OPERATING SYSTEMS**

SECTION : **SECR/J/B/V/P/ SCSR/J/B/V/P SEEL/H**

TIME :

DATE/DAY :

VENUES :

INSTRUCTIONS:

Answer all questions from PART A, and PAPRT B. Show all your works.

This test will contribute **35%** towards the total marks of 100%.

(Please Write Your Lecturer Name and Section)

Name	
I/C No.	
Year / Course	
Section / Lecturer Name (Circle)	**[01] - Dr Farkhana** **[02] - Dr Syed Hamid Hussain Madni** **[03] - Dr Syed Hamid Hussain Madni** **[04] - Dr Zuriahati** **[05] - Dr Farkhana** **[07] – Dr Zuriahati**

This question booklet consists of 12 pages including the front page.

PART A: OBJECTIVES

Answer ALL questions.

1. To ensure that the circular wait condition would not happen, a deadlock avoidance algorithm directly evaluates the _____.

A. Mutual exclusion, no-preemption, hold and wait, and circular wait.
B. Blocking send, race condition, cache incoherency and RAM overflow.
C. Mutual exclusion, race condition, cache incoherency, and RAM overflow.
D. Blocking send, race condition, hold and wait, and RAM overflow.

2. Which of the following statements concerning deadlock recovery is **FALSE**?

A. Abort all deadlock processes.
B. Abort one process at a time until the deadlock cycle is eliminated.
C. Rollback - return to some safe state, restart process for that state.
D. Banker's algorithm will help to recover.

3. To ensure that deadlock never occurs, the system can use _____.

I. deadlock prevention scheme
II. deadlock avoidance scheme
III. deadlock detection scheme
IV. deadlock recovery scheme

A. I & II
B. II & III
C. I & IV
D. II & IV

4. An address generated by the CPU is commonly referred to as a _____.

A. physical address.
B. space address.
C. virtual address.
D. memory address.

5. A process can be swapped temporarily out of memory to a _____ and then brought back into memory for continued execution.

A. register
B. backing store
C. processor
D. buffer

6. Given Table 1, what are the physical address for the logical address of 3:400?

Table 1

Segment	Base	Length
3	1327	580

A. 1727
B. 980
C. 1327
D. 927

7. Which of the following is the **TRUE** sequential order of storage devices from fastest to slowest?

A. Solid State, Optical disk, Magnetic disk
B. Optical disk, Magnetic disk, Solid State
C. Magnetic disk, Optical disk, Solid State
D. Solid State, Magnetic disk, Optical disk

8. The technique of duplicating every disk is known as:

A. Mirroring
B. Scheduling
C. Redundancy
D. Shadowing

9. RAID 3 does not have _____ as like in RAID 1.

A. Storage Overhead
B. Enough Space
C. Less Space
D. Time Consumption Overhead

10. Which of the following statements about file-system consistency is **TRUE**?
A. It can always be ensured even in the case of system crash.
B. It is not affected by system crashes.
C. It may be compromised in the event of a system crash.
D. None of the above.

11. The file allocation methods that require an external table for keeping track of used and unused blocks is _____.
A. Contiguous allocation
B. Linked allocation
C. Indexed allocation
D. None of the above

12. What is the main advantage of a linked-list free-space management system over a bit map?
A. It requires less memory to store the free-space list.
B. It provides faster access to the free-space list.
C. It can compact the free space more easily.
D. It finds the first free block of sufficient size more quickly.

13. Which action will not generate the page fault?
A. Find a free frame in memory.
B. Read a page that is already in the memory.
C. Restart the instruction.
D. Swap the page into frame via schedule disk operation.

14. Choose a page-replacement algorithm that suffer from Belady's anomaly.
A. Optimal replacement

B. LRU replacement

C. FIFO replacement

D. Round Robin replacement

15. Based on **Figure 1**, what will happen if the process is requested for Page B? Assume that the backing store contains only Pages D, E, G and H.

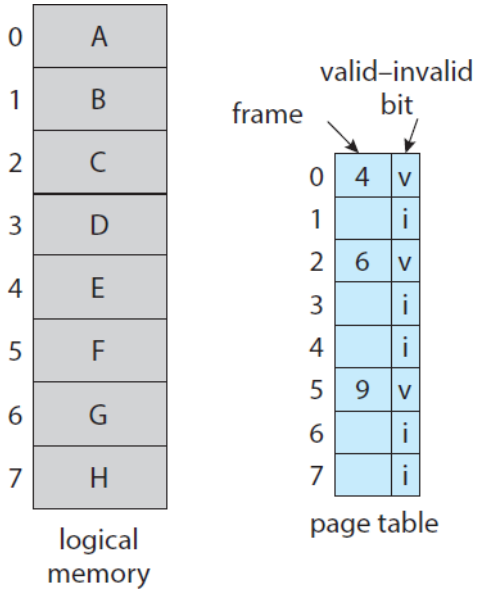


Figure 1

A. The process will be terminated.

B. Page B will go through a swapping process.

C. The process can directly be executed.

D. The process will go into the waiting state.

16. Swapping is introduced to address the drawbacks of virtual memory, what is the effect of bringing a page into memory only when needed?

A. Slower response.

B. Reduce the number of users.

C. Less I/O is needed.

D. More memory is needed.

17. How is a trap generated during demand paging?

A. When OS fails to bring the designed page into the memory.

B. When there is no free space in the main memory.

C. When OS reset the page table to valid.

D. When the page is fetched from the backing store into the main memory.

18. What job will be carried out in the file-organization module?

A. Issue the general commands to the device driver.

B. Manage the metadata information.

C. Information is transfer between the main memory and the disk system.

D. Translate the logical block number to physical block number.

19. Which statement about a file system is **FALSE**?

A. Reside on secondary storage.

B. Provide user interface to storage, mapping logical to physical.

C. Manage organization and storage of files on a computer.

D. Provide efficient and convenient access to disk.

20. If no frames are free, _____ page transfer(s) is/are required.

A. one

B. two

C. three

D. four

PART B: STRUCTURED

Answer ALL questions in the answer booklet.

QUESTION 1 [18 MARKS]

(a) Given seven dynamic partition memory at time t_0 as shown in **Figure 2**.

Assume Job *P* and *Q* finished at t_1, and the new jobs arrived in order at t_3 Job *T* (30 *KB*) and Job *U* (15 *KB*), and at t_5 Job *V* (20 *KB*).

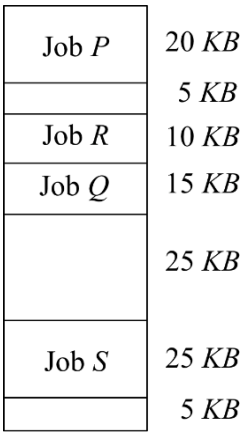


Figure 2

Redraw **Figure 2** with all allocated jobs in the memory at t_1, t_3, and t_5, respectively. [6]

(b) Given six memory partitions, as shown in **Figure 3**.



Figure 3

Consider five processes (P_1, P_2, P_3, P_4, P_5) with different sizes need to place in the memory (in order) as shown in **Table 2(a)** using the first-fit, best-fit, and worst-fit algorithms. Reconstruct and complete **Table 2(b)** by writing the **memory partition numbers** for each algorithm, respectively. [4]

Table 2(a)

Processes	Size (KB)
P_1	115
P_2	500
P_3	355
P_4	200
P_5	375

Table 2(b)

(Copy the table to answer booklet)

	First-Fit	**Best-Fit**	**Next-Fit**	**Worst-Fit**
i)				
ii)				
iii)				
iv)				
v)				

(c) Consider a paging system for a logical address space of 2^4 pages (bytes) with a page size of 2^2 bytes and a physical memory of 64 bytes. Answer the following questions. Show your working.

i. What is the frame size of the physical memory? [2]

ii. How many frames of the physical memory? [2]

iii. How many number of entry in the page table? [2]

iv. Assume the frame number for a data is 8 with offset 1. Calculate the physical address for the data. [2]

QUESTION 2 [19 MARKS]

(a) Given a set of vertices, edges, and process states.

P = {P1, P2, P3, P4}

R = {2R1, 2R2, 2R3, R4}

E = {P1 ⇨ R2, P2 ⇨ R3, P3 ⇨ R2, P3 ⇨ R4, P4 ⇨ R4, R1 ⇨ P1, R1 ⇨ P2, R2 ⇨ P2, R3 ⇨ P3, R3 ⇨ P4}

i. Draw the complete RAG. [4]

ii. Is there any deadlock? [2]

(b) Given the following resource allocation in **Table 3**.

Table 3

	Allocation			Max			Available		
	A	B	C	A	B	C	A	B	C
P1	1	2	0	5	5	3	2	4	3
P2	1	1	0	2	3	3			
P3	0	1	3	3	3	6			
P4	3	1	0	5	7	2			
P5	1	0	1	2	0	2			

i. What is the total number of each resource in the system? [2]

ii. Compute the Need Matrix based on Safety Algorithm. [3]

iii. Is the system in a safe state? Justify your answer. [4]

iv. Assume P1 requests (2, 2, 0) as shown in Table 2, and determine by using the Safety algorithm whether the operating system should meet P1's request. Show all your work. [4]

QUESTION 3 [16 MARKS]

Following is the disk scheduling algorithms information:

- A disk drive with 250 cylinders (0-249).
- A request queue (98, 200, 37, 125, 15, 100, 26, 217).
- Head pointer at 50.

(a) Redraw the movement of the disk arm and calculate the total distance (in cylinders) to satisfy the requests. Show your work.

i. SSTF. [5]
ii. C-LOOK (Innermost) . [5]
iii. SCAN (Innermost) . [5]

(b) From **Questions (a)(i– iii)** choose the best algorithm that performs optimally in terms of disk scheduling? [1]

QUESTION 4 [12 MARKS]

The CPU accesses the addresses based on the following reference string:

i. Identify the total number of page fault using Optimal Page Replacement (OPT) and 5 frames. Show your working. [5]

ii. Identify the total number of page fault using Least Recently Used (LRU) and 5 frames. Show your working. [5]

iii. From Question 2(i) and 2(ii), choose the best replacement technique state **ONE** of the drawbacks of the chosen technique. [2]

QUESTION 5 [15 MARKS]

(a) **Figure 4** shows a bit map for free space management for a hypothetical Operating System whereby bit 0 indicates that the block is used, and bit 1 indicates that the block is free. Suppose that the size of a block is 15KB.

Block Number	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Bit Map	0	1	1	0	1	0	0	1	0	1	1	1	0	1	0	0

Figure 4

i) If the size of the DATA.txt is 84 KB, list all the block numbers that will be used to allocate the file. Show your work. [2]

ii) Based on the answer in (a)(i), is there any fragmentation occurred? If yes, calculate the size of fragmentation by showing your workings. If no, explain your answer briefly. [2]

(b) A system has a disk with 20 storage blocks, with the size of each storage block is 15KB, as shown in **Figure 5**.

1	19	2	20	3	11	4	14
5	-	6	16	7	15	8	18
9	4	10	6	11	17	12	-
13	3	14	-	15	9	16	7
17	1	18	5	19	-	20	8

Figure 5

Assume a files-linked allocation is used for files A and B, which start at blocks 2 and 13, respectively. Answer the following questions.

i) Write all blocks used by each file. [2]

ii) Calculate the total size of each file. Assume that there is no fragmentation occurred. Show your working. [2]

(c) Consider that a file system needs to manage the following free-space lists to easily track available blocks in the future.

3, 16, 7, 19, 4, 5, 9, 12, 2, 10, 15, 8, 20, 1, 6, 11, 14, 23

Assume each block can only store **five** block addresses with a grouping scheme. Answer the following questions.

i) List all the index blocks. [2]

ii) Draw all the blocks used. [5]